

AI Engineering Career Roadmap

Student → AI / Systems Engineer · 6-Month Sprint · Built with Chain-of-Thought Reasoning

CURRENT POSITION

Student — CS / Engineering background

- Python & Data manipulation
- Java, Spring Boot, Microservices
- Docker containerisation
- AWS cloud fundamentals

TARGET GOAL

Full-Time AI / Systems Engineer

- Build scalable, production-grade AI systems
- Master LLMs, RAG, inference optimisation
- Core systems: distributed computing, low-latency
- High-impact role at top-tier tech company

SKILL GAP ANALYSIS

Area	Current Skills	Target Skills	Priority
AI / ML Core	Python, basic NumPy	PyTorch, Transformers, fine-tuning, RAG	■ Critical
LLM Engineering	—	LangChain/LlamaIndex, prompt eng, vector DBs	■ Critical
Systems Design	Microservices, Docker, AWS	Distributed systems, gRPC, Kafka, Redis	■ High
ML Infra / MLOps	Docker, AWS basics	Kubernetes, model serving, CI/CD for ML	■ High
Data Engineering	Python	Spark, Airflow, data pipelines at scale	■ Medium
Algorithms / DSA	Java, Python	LeetCode Medium/Hard, system design interviews	■ Critical

LEARNING PLAN

Month 1–2 · AI Foundations

- Fast.ai / Andrej Karpathy NanoGPT series
- HuggingFace Transformers course (free)
- PyTorch fundamentals + backprop from scratch

Month 2–3 · LLM Engineering

- RAG pipelines: LangChain + FAISS / Pinecone
- Fine-tuning: LoRA/QLoRA on open models
- Prompt engineering & evaluation frameworks

Month 3–4 · Systems + MLOps

- gRPC, Kafka, Redis — distributed patterns
- Kubernetes + model serving (Triton / vLLM)
- CI/CD pipelines for ML (GitHub Actions)

Month 4–6 · Interview Prep (parallel)

- LeetCode 150 (Medium/Hard) — daily 1–2 Qs
- System design: Grokking the System Design
- Mock interviews — Pramp / Interviewing.io

SUGGESTED PROJECTS

1. Mini-GPT from scratch

Implement transformer in PyTorch; train on tiny-shakespeare. Proves deep understanding of LLMs.

2. RAG-powered Document QA API

LangChain + FAISS + FastAPI + Docker. Deploy on AWS ECS. Shows end-to-end LLM engineering.

3. Distributed Inference Server

Serve an open LLM with vLLM behind gRPC gateway; load test with Locust. Kafka for async jobs.

4. Fine-tuned Code Assistant

LoRA fine-tune CodeLlama on GitHub dataset; build VS Code extension. Killer portfolio piece.

5. ML Pipeline on Kubernetes

Airflow DAG → model training → canary deploy with Argo Rollouts. Demonstrates MLOps maturity.

NETWORKING STRATEGY

- LinkedIn:** Post weekly build-logs, project demos & learnings
- GitHub:** All 5 projects public; clean READMEs with demos
- Twitter/X:** Engage with AI researchers & open-source devs
- Communities:** Hugging Face, LangChain Discord, MLOps Community
- Cold outreach:** 3 engineers/week via LinkedIn DM — ask for 15-min calls
- Conferences:** NeurIPS workshops, AI Engineer Summit (virtual)
- Open Source:** Contribute 1 PR/month to HuggingFace or LangChain

MONTHLY MILESTONES

Month 1	PyTorch fluent · Mini-GPT trained & on GitHub
Month 2	RAG API live on AWS · 50 LeetCode Mediums done
Month 3	Distributed inference server deployed · LoRA fine-tune done
Month 4	ML K8s pipeline live · 100 LeetCode · 5 LinkedIn posts
Month 5	All 5 projects polished · 10+ interviews running
Month 6	Offer signed · negotiate salary confidently ■

IMMEDIATE NEXT ACTIONS — DO THIS WEEK

■ Set up PyTorch environment + run first neural net

■ Start Karpathy's 'makemore' lecture series tonight

■ Publish GitHub profile README with tech stack

■ Solve 2 LeetCode Mediums (Arrays / Sliding Window)